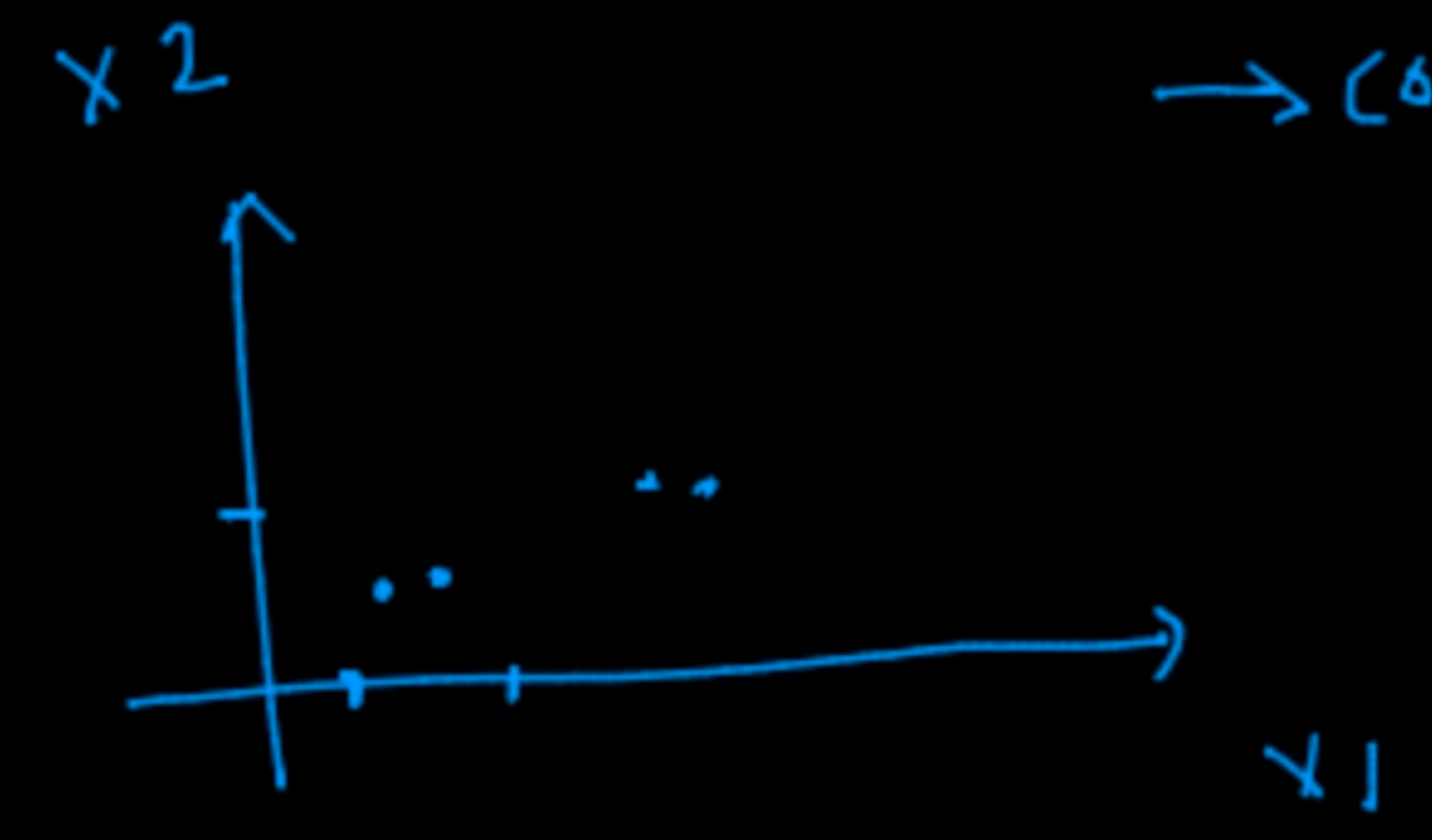


Linear Algebra

SL	pw	
X1	X2	Target
(3	2)	Yes
(1	2)	Yes
(4	5)	No
(6	5)	NO



lose less
↑
'distance'

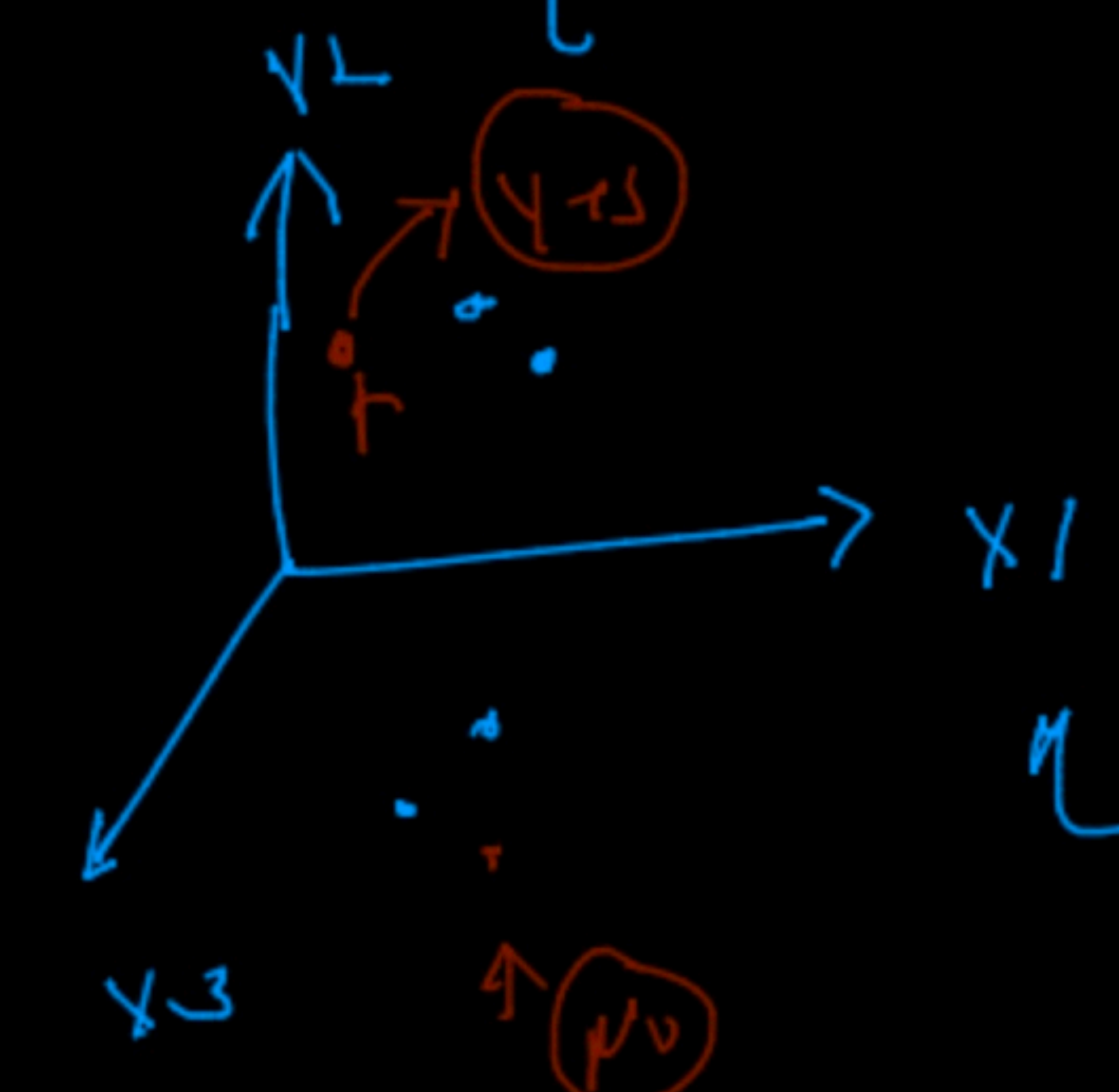
if $x1 < \text{average}$ → Yes

else

if $(x1 \text{ around } x2)$ → Yes/No

X	Target
1 3	Yes
2 2	Yes
3 1	No
4 4	NO
5 6	
6 2	
7 3	

plot →

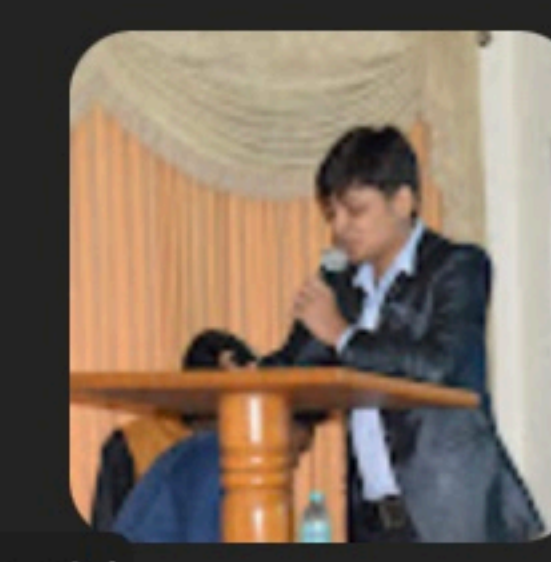


Rule

this dataset

(4D)

Yes/NO



Rahul Tripathi



Akila Anbazhagan

Handwritten notes on the left side of the screen:

x_1	x_2	x_3	x_4	Result
3	2	1	6	Yes
6	5	2	1	No

→ plus x

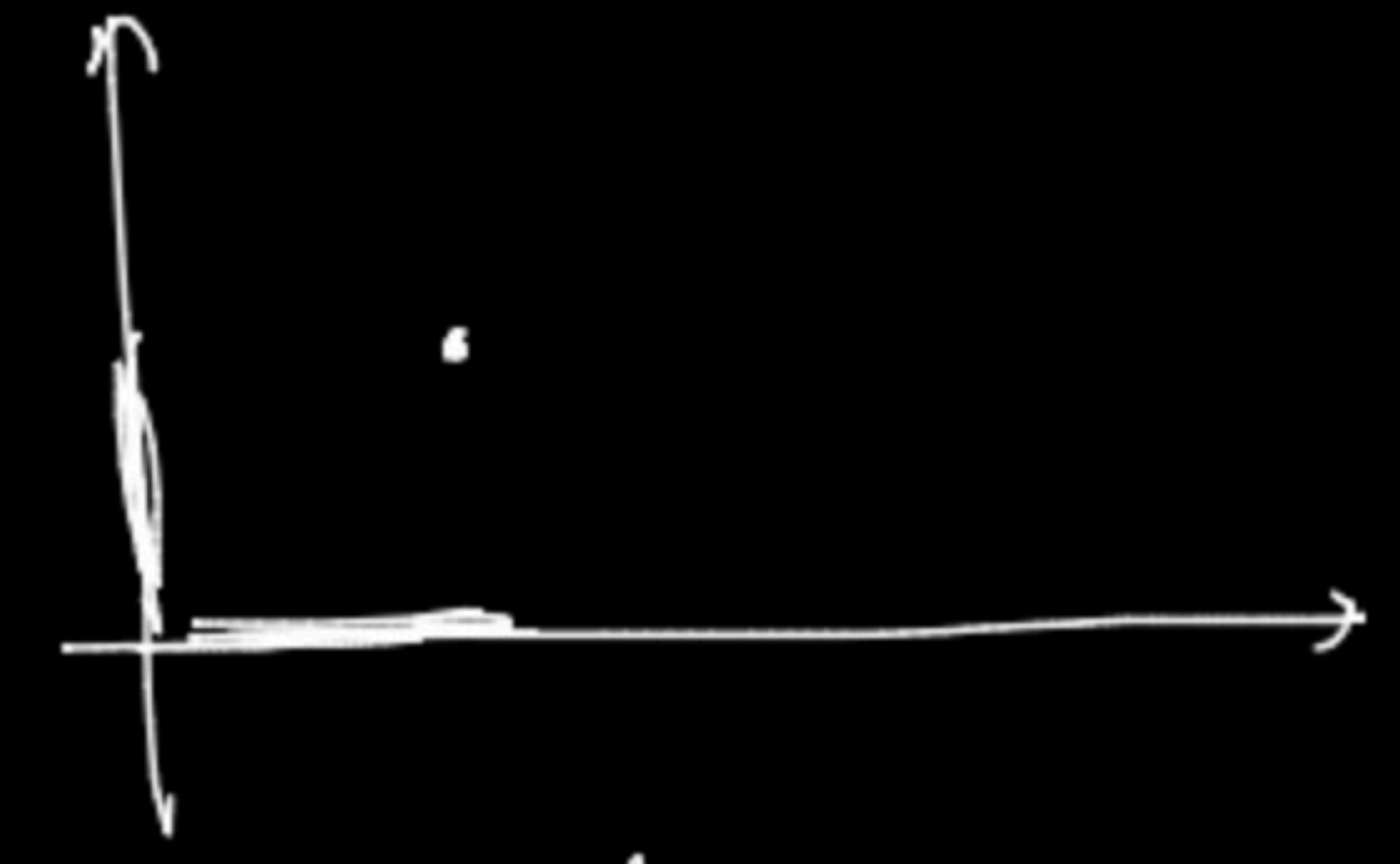
Linear Algebra → math

$D_x \rightarrow LA \rightarrow conclusion$

Handwritten notes on the right side of the screen:

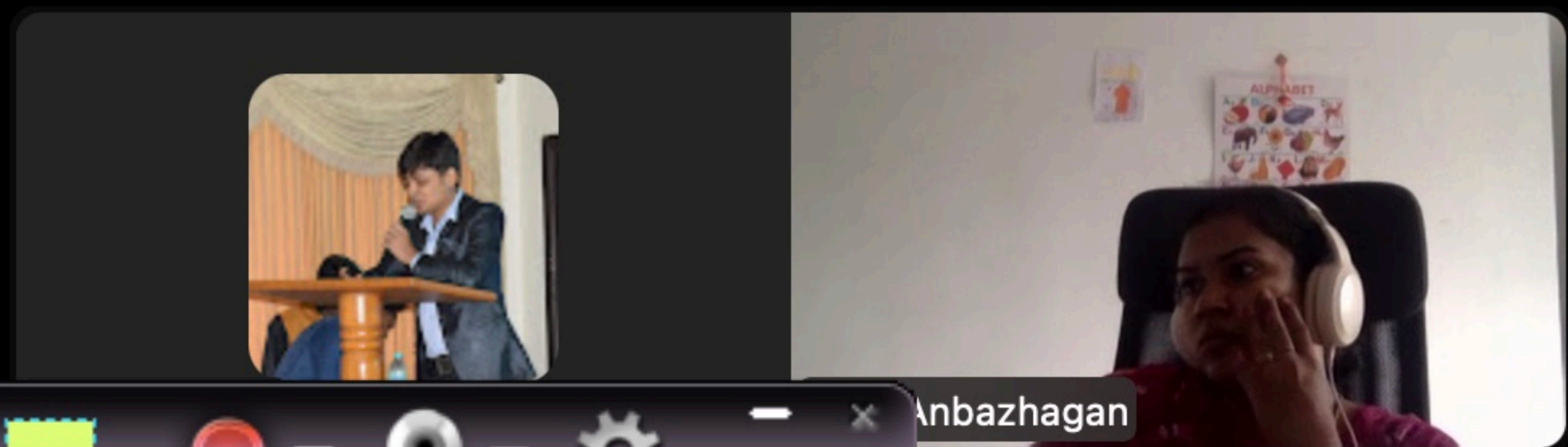
① Points on vector x_1

$P(x_1, x_2)$ i.e. $(2, 3)$



$\uparrow$ 4D, 5D - $nD?$

You are screen sharing Stop share

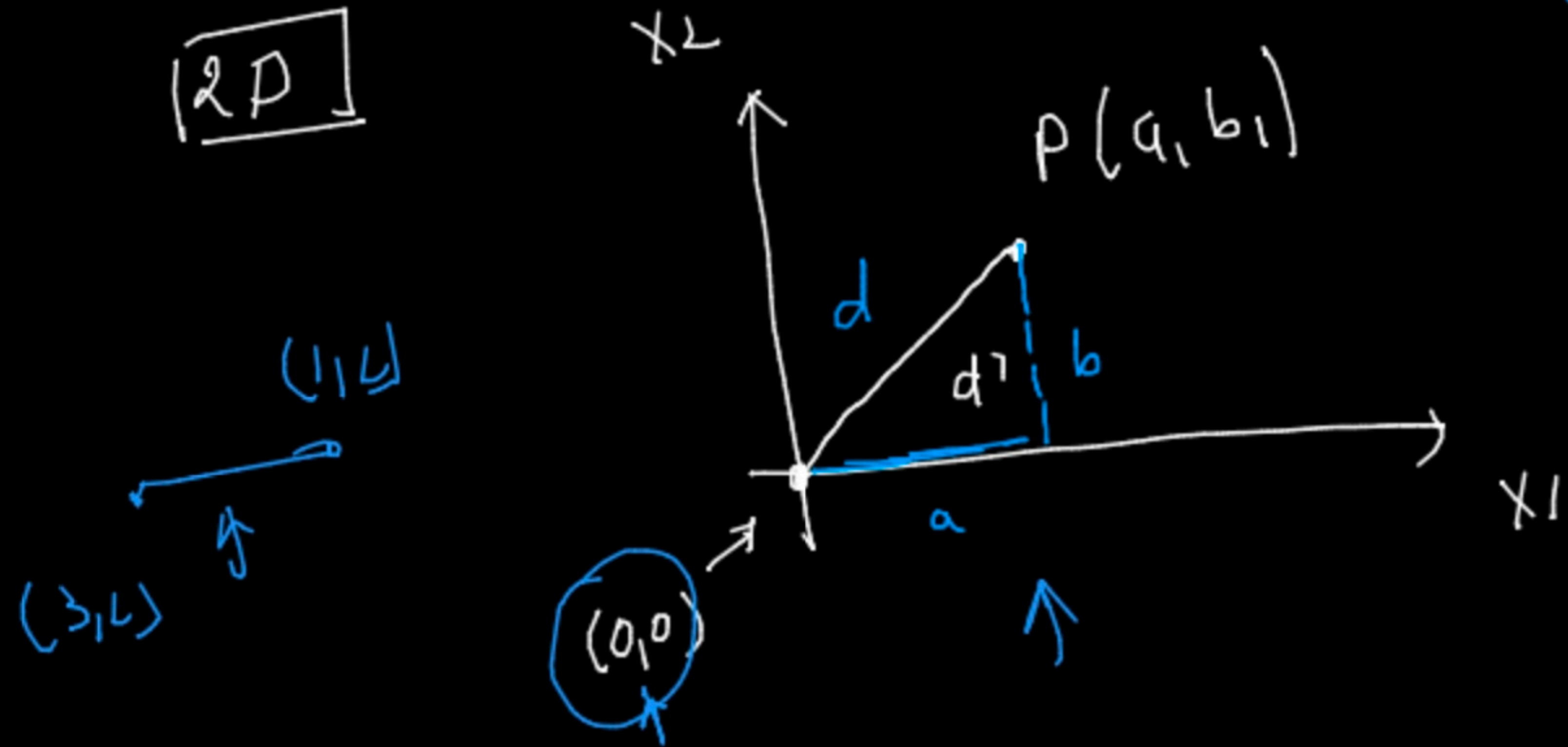


Handwritten notes at the bottom of the screen:

2 / 2

② Distance of a point from origin

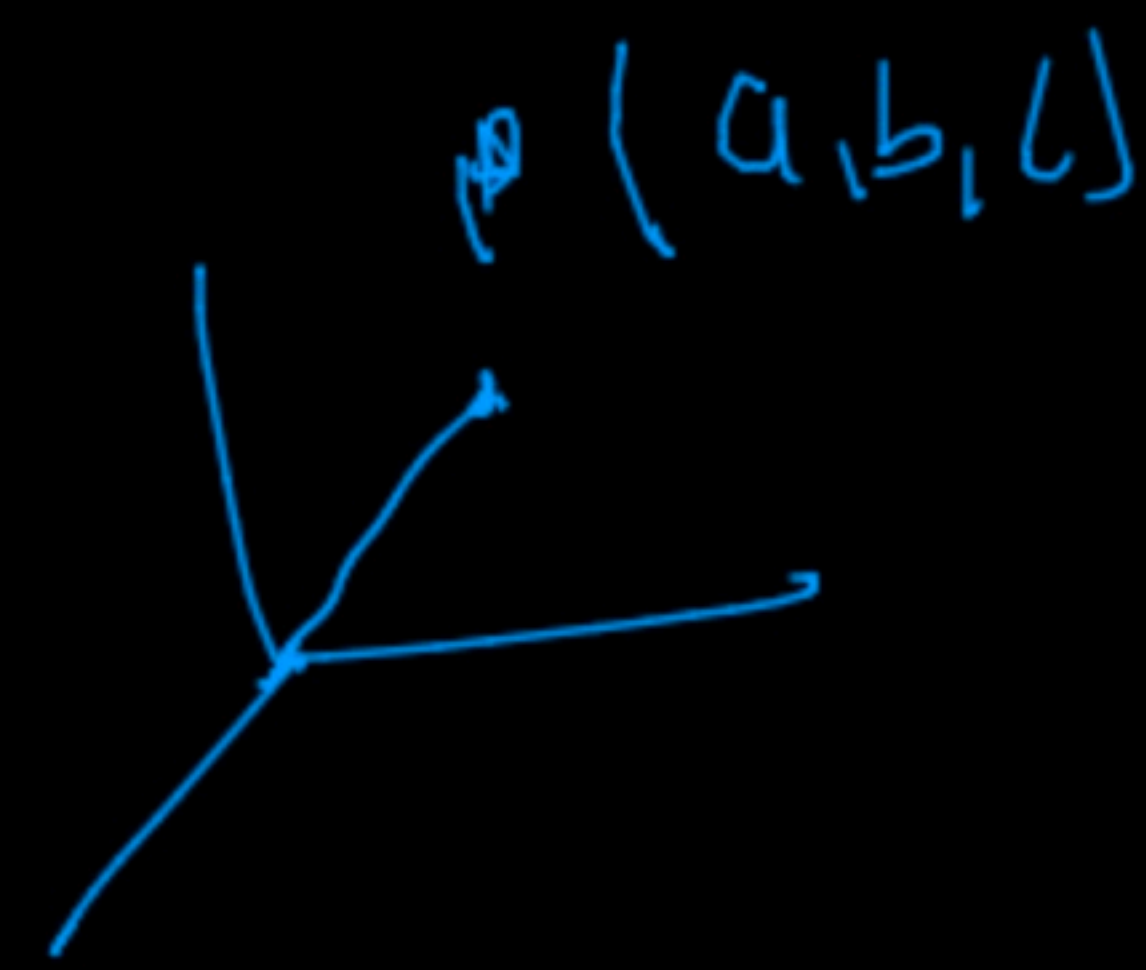
(2D)



$$d^2 = a^2 + b^2$$

$$d = \sqrt{a^2 + b^2}$$

(3D)

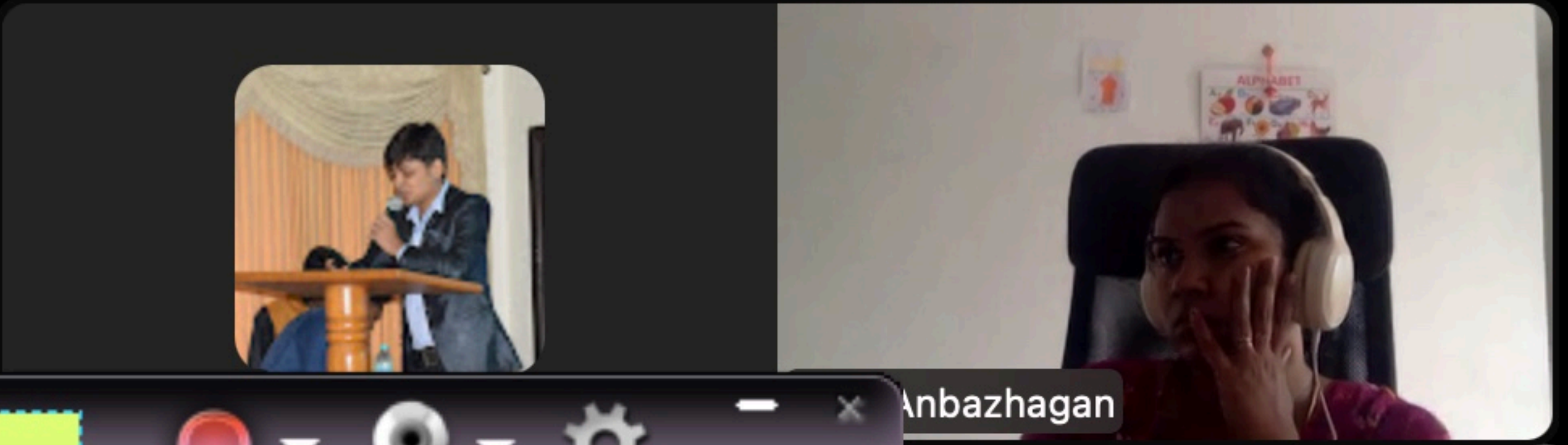


$$d = \sqrt{a^2 + b^2 + c^2}$$

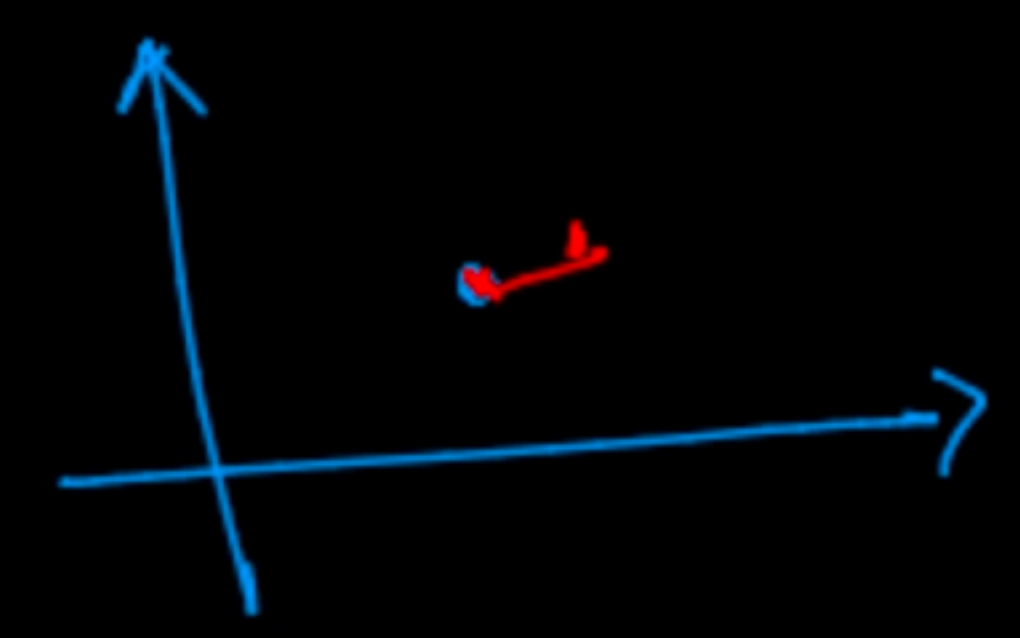
nd - $P(a_1, a_2, a_3, \dots, a_n)$

$$d = \sqrt{a_1^2 + a_2^2 + a_3^2 + \dots + a_n^2}$$

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$C1 = (14, 750)$
 $C2 = (14, 750)$ → 0 → exactly same
 $C3 \rightarrow [17, 780]$



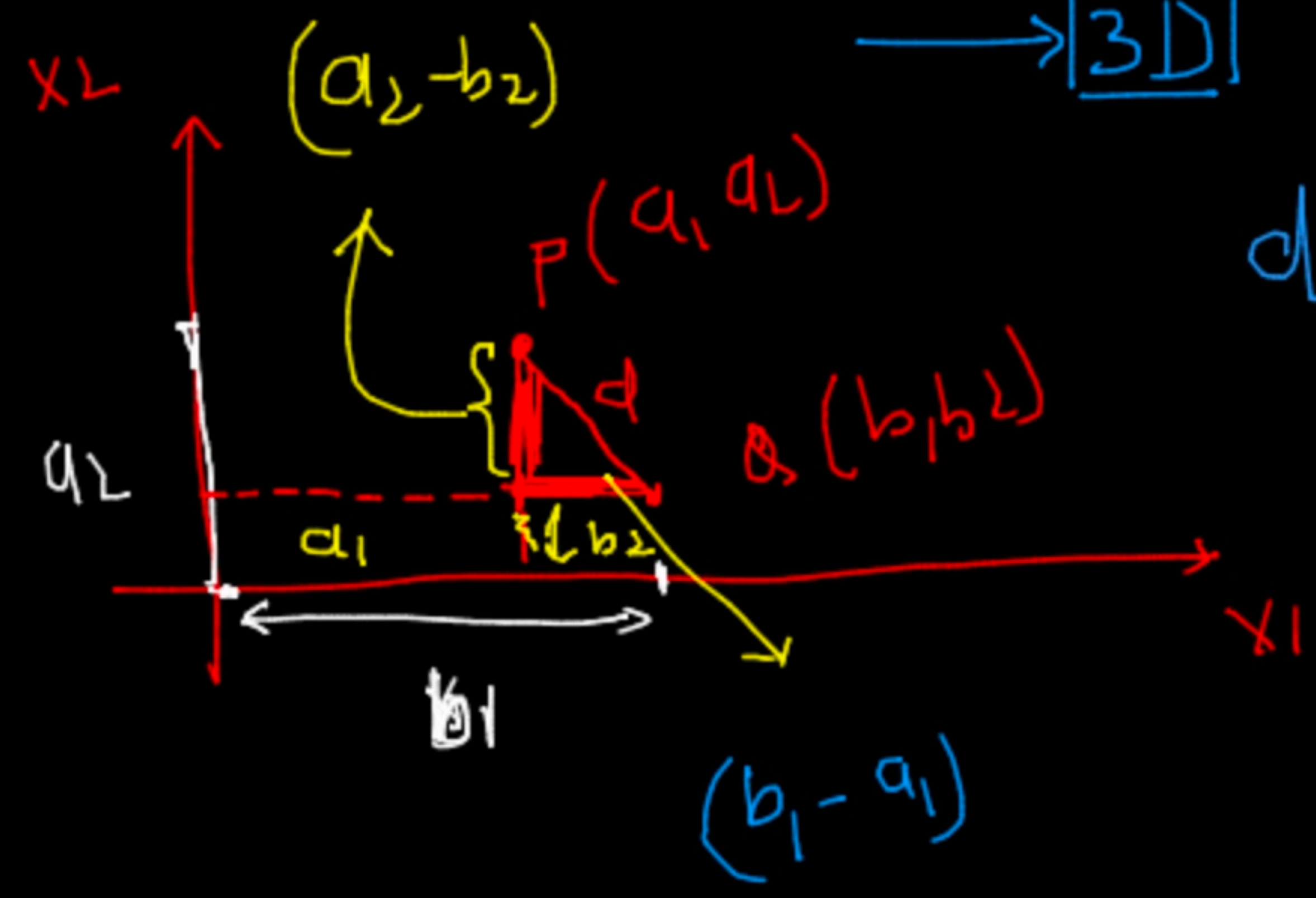
$P(a_1, a_2, a_3)$
 $Q(b_1, b_2, b_3)$

distance b/w 2 points

$(7, 780)$ } use $C1$
 } $C2$
2D

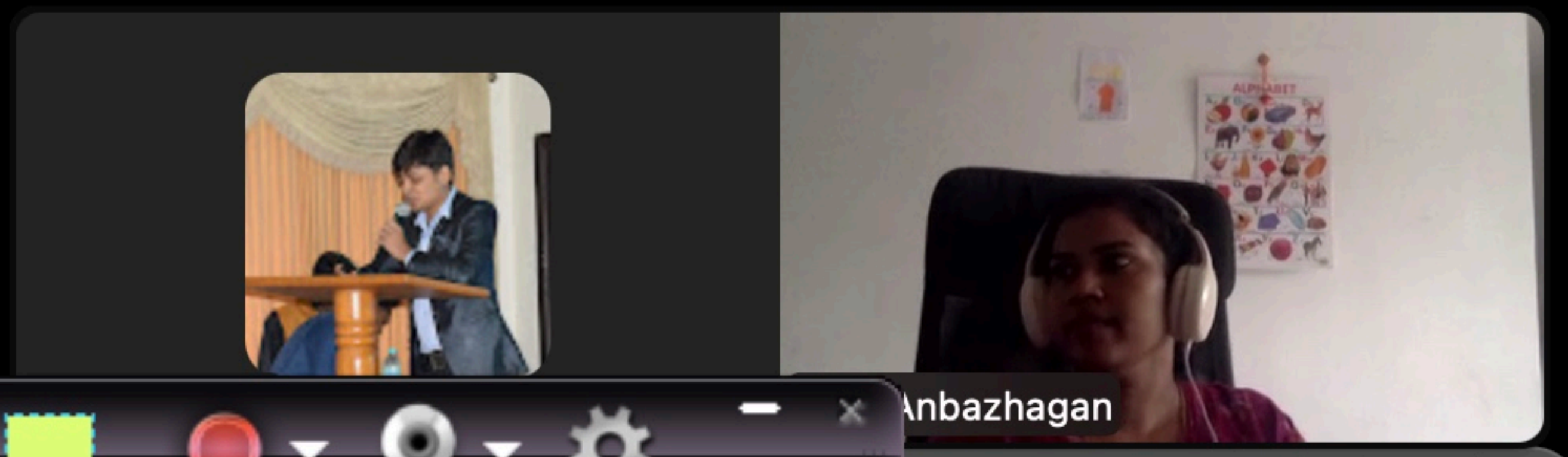
→ 3D

$$d = \sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2}$$



$$d = \sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2 + (a_3 - b_3)^2}$$

You are screen sharing Stop share



np $\rightarrow$ $p(a_1 a_2 a_3 a_4 \dots a_n)$
 $Q(b_1 b_2 b_3 b_4 \dots b_n)$

Euclidean

Note $\rightarrow$ $d = \sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2 + \dots + (a_n - b_n)^2}$

$\downarrow$
code $\rightarrow$
 $d = \text{sqrt}\left(\sum_{i=1}^n (a_i - b_i)^2\right) \leftarrow$
 $\uparrow$
input $\nwarrow$ numpy

You are screen sharing Stop share

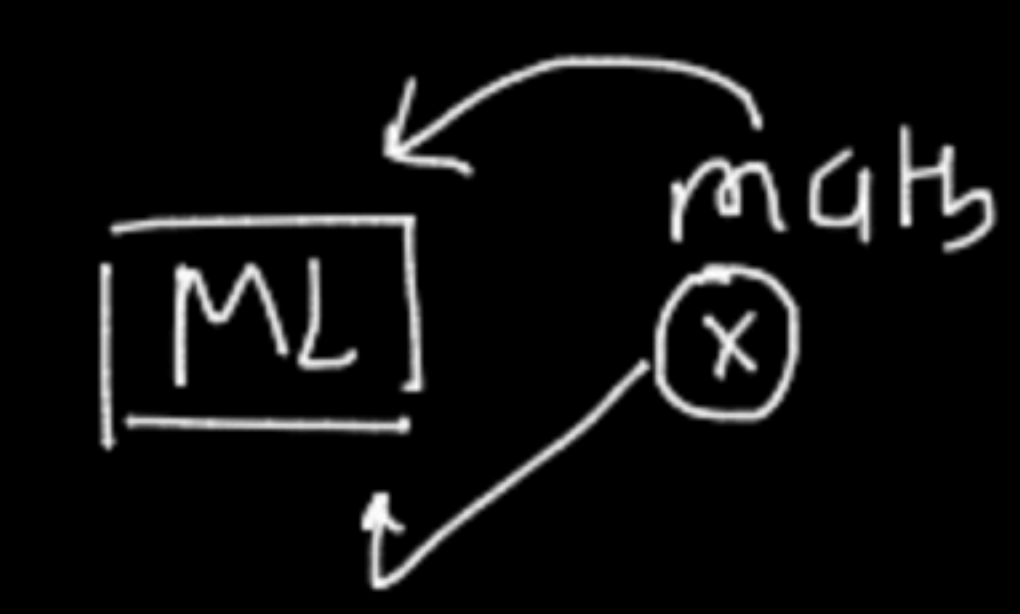


✓ ① Row vector
 ✓ ② Col vector

} → operation

→ $[a_1, a_2, a_3, a_4, a_5, \dots, a_n]$

① $a + b$



$$\begin{aligned}
 a &= [a_1, a_2, a_3, \dots, a_n] \\
 b &= [b_1, b_2, b_3, \dots, b_n]
 \end{aligned}$$

$\uparrow \quad \uparrow \quad \uparrow$

② $a \times b$

→ Dot product ✓
 → Cross product → $[ML] \times$

$a \cdot b = (1 \times n) (1 \times n)$

↓
 same $[X]$

$a_1 \cdot b_1 + a_2 \cdot b_2 + a_3 \cdot b_3 + \dots + a_n \cdot b_n$

$\downarrow$

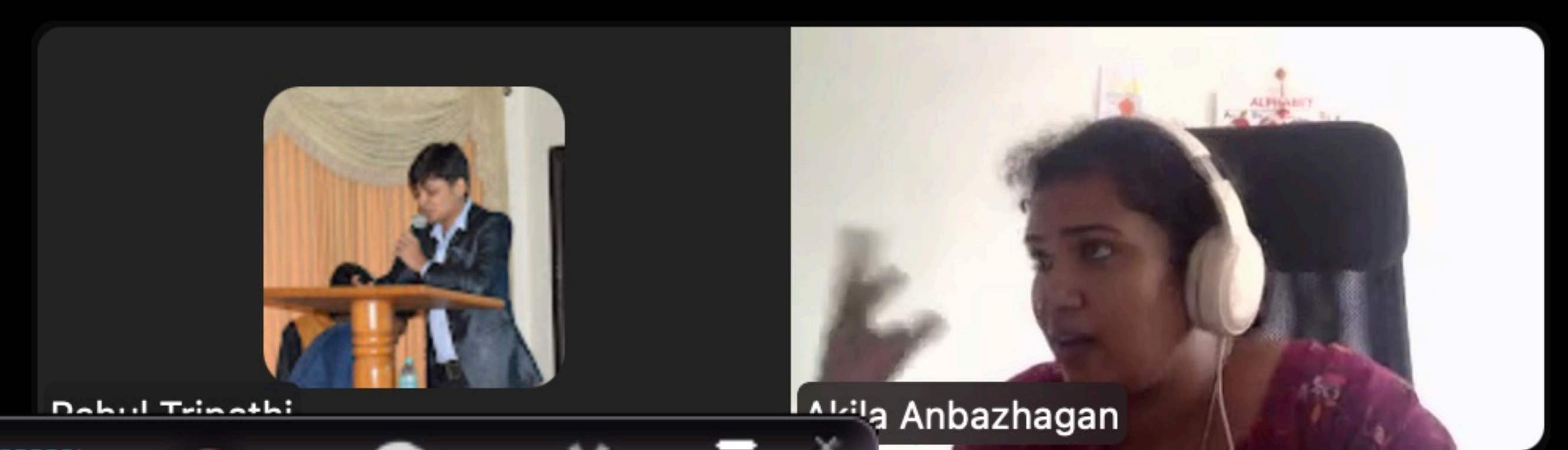
$\sum_{i=1}^n (a_i \cdot b_i)$

$\left[\begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} \right] \left[\begin{pmatrix} b_1, b_2, \dots, b_n \end{pmatrix} \right]$

$a \cdot b = a^T \cdot b$

$= (n \times 1) \times (1 \times n)$

You are screen sharing Stop share



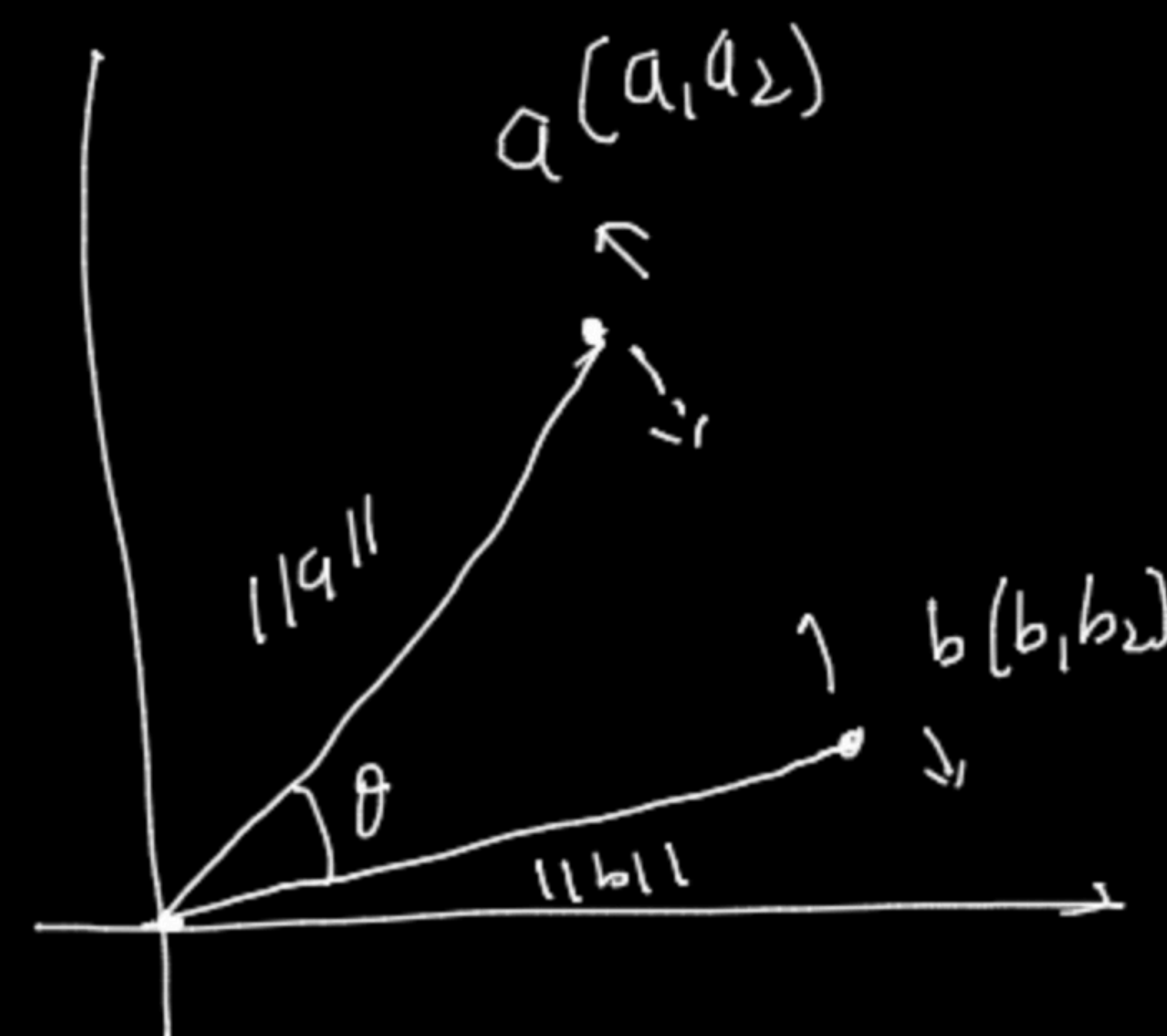
dot $\Rightarrow$

$$\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} [b_1, b_2] \quad \boxed{ML} \quad \Rightarrow$$

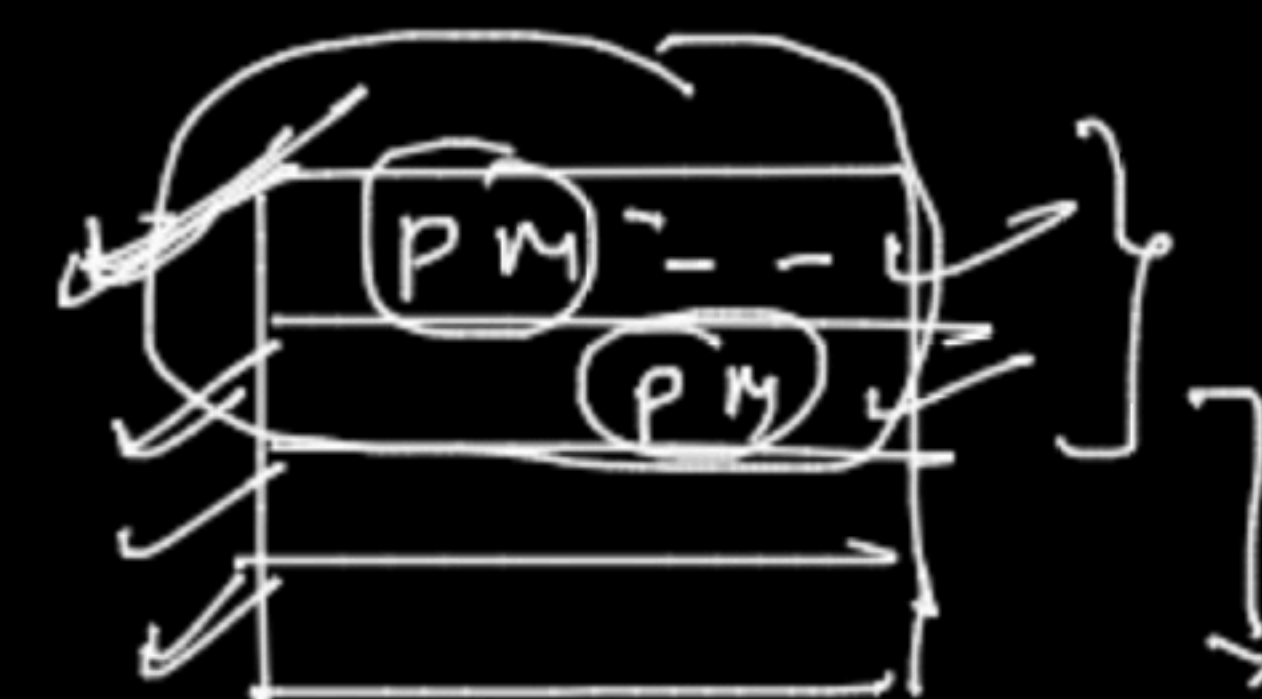
$$a_1 \cdot b_1 + a_2 \cdot b_2$$

$$\sum_{i=1}^n a_i^T \cdot b_i$$

dot $\Rightarrow$ $[---]$

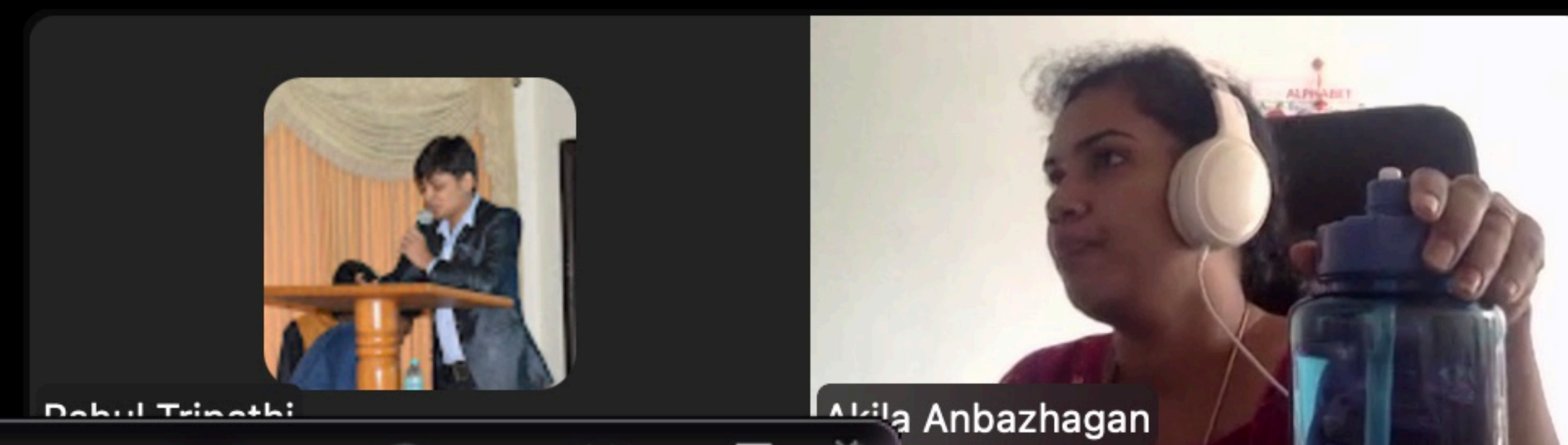


$\theta \downarrow \uparrow$



$\Rightarrow$ $\theta \downarrow$ Its $\rightarrow$ similar $\Rightarrow$ $[---]$
 $\uparrow$
cosine-similarity
(text) $\rightarrow$ "who is PM of INDIA"

You are screen sharing Stop share



$$a \cdot b = \|a\| \|b\| \cos \theta$$

You are screen sharing

Stop share

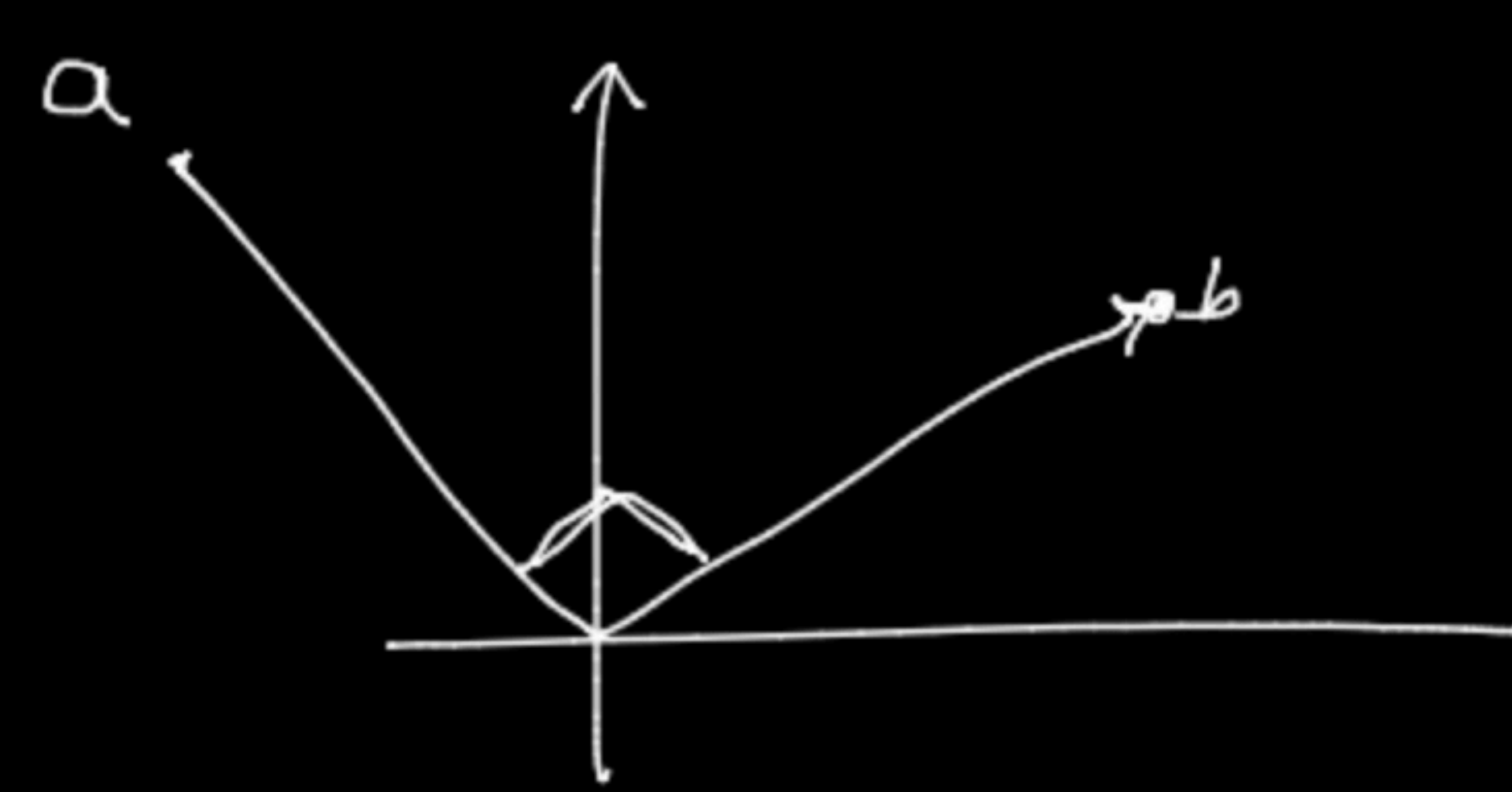
$$a_1 \cdot b_1 + a_2 \cdot b_2 = \|a\| \|b\| \cos \theta$$

$$\frac{a_1 \cdot b_1 + a_2 \cdot b_2}{\|a\| \|b\|} = \cos \theta$$

$$\theta = \cos^{-1} \left(\frac{a_1 \cdot b_1 + a_2 \cdot b_2}{\|a\| \|b\|} \right)$$

2D

$$\theta = \cos^{-1} \left(\frac{\sum_{i=1}^n a_i \cdot b_i}{\|a\| \|b\|} \right)$$



diso !

$$a \cdot b = 0$$

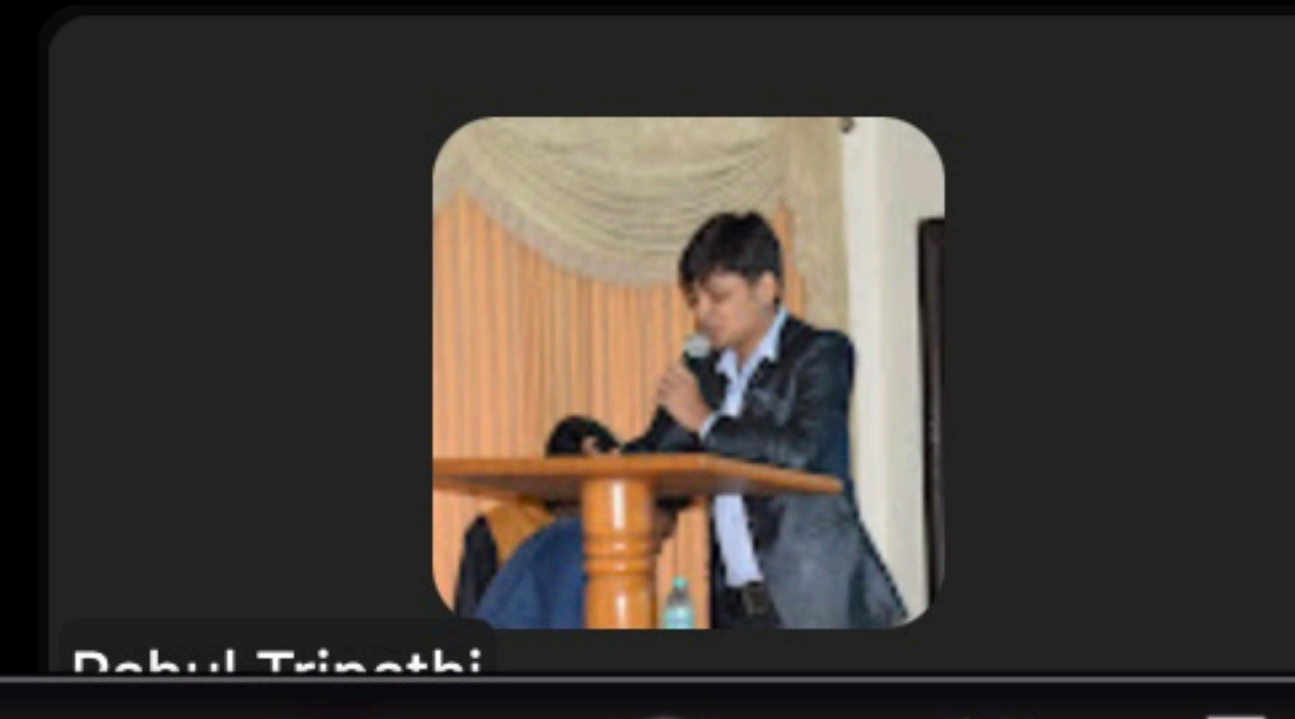
$$\|a\| \cdot \|b\| \cos \theta = 0$$

||a||



(lat, lng) haversine formula

→ 'haversine'



Rahul Tripathi



Alila Anbazhagan

Line

$y = mx + c$
← slope ← Intercept

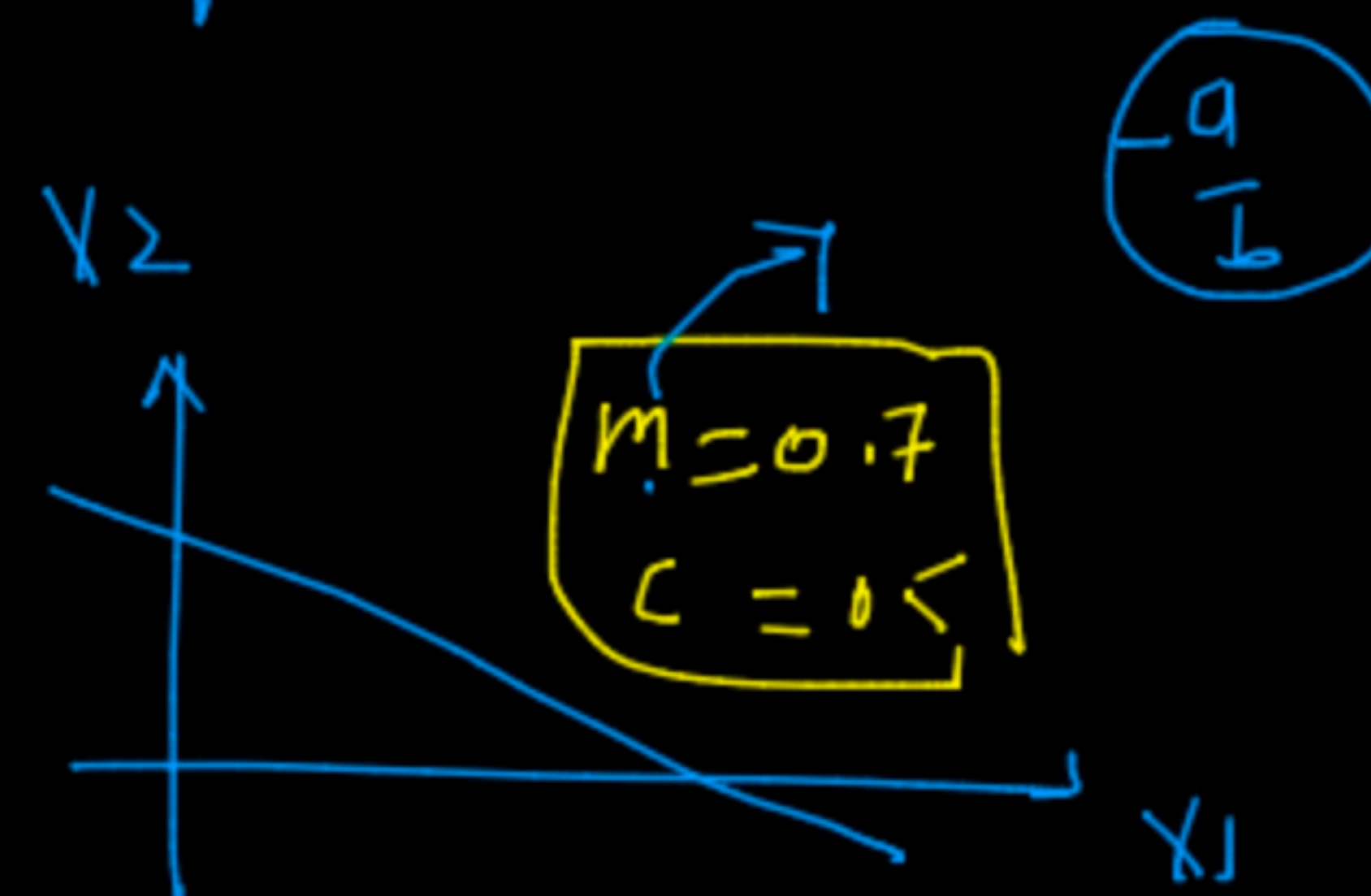
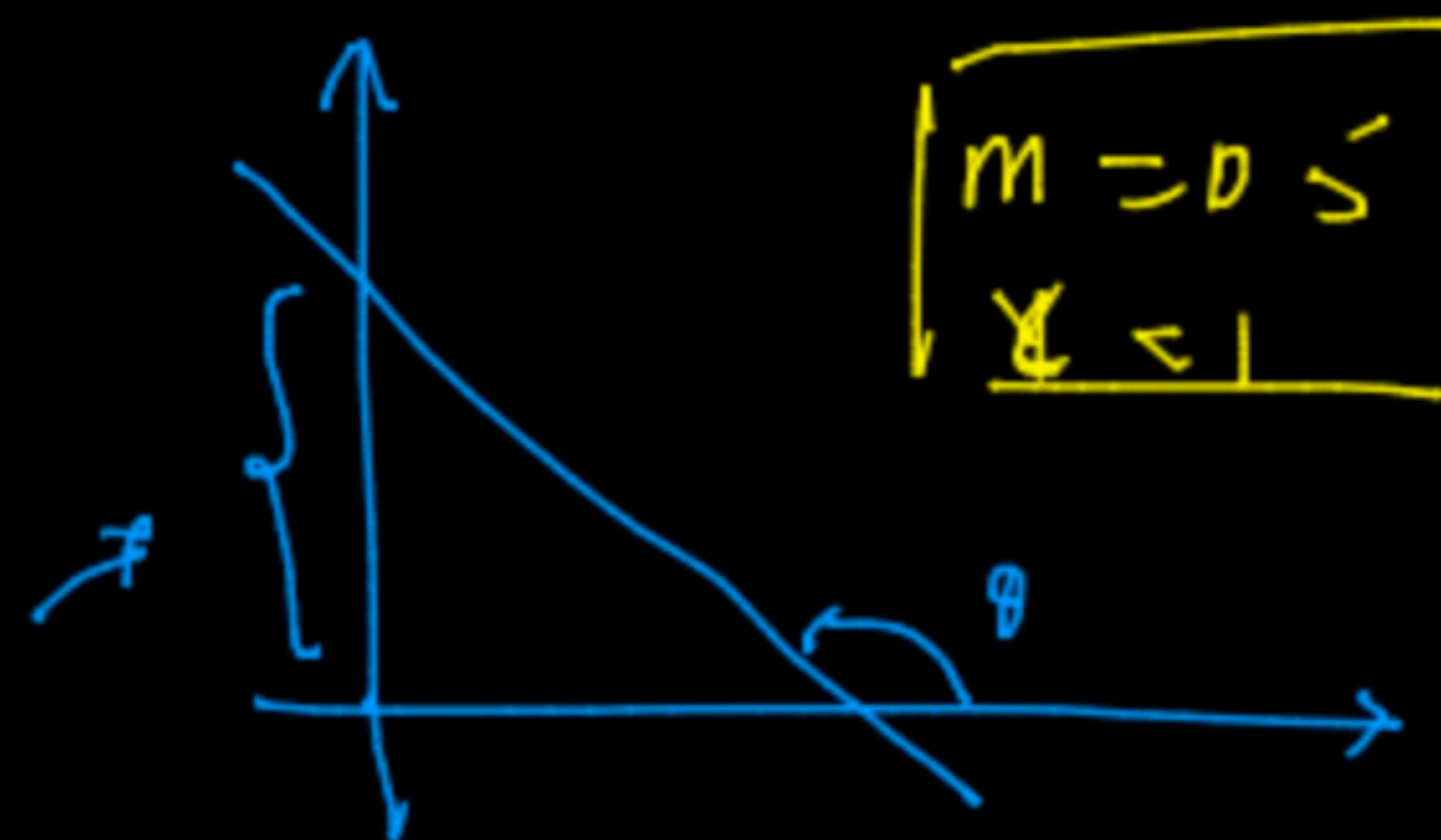
↑
General equation of line
y-intercept

$ax + by + c = 0$

↑
 $by = -ax - c$

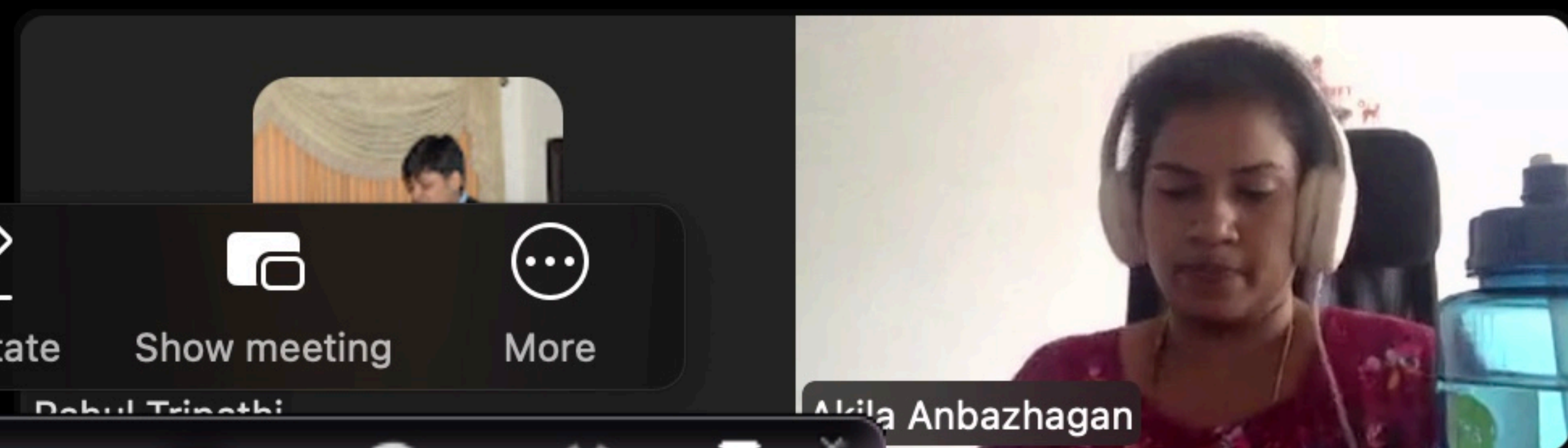
$y = -\frac{a}{b}x - \left(\frac{c}{b}\right)$

⇓



eg.

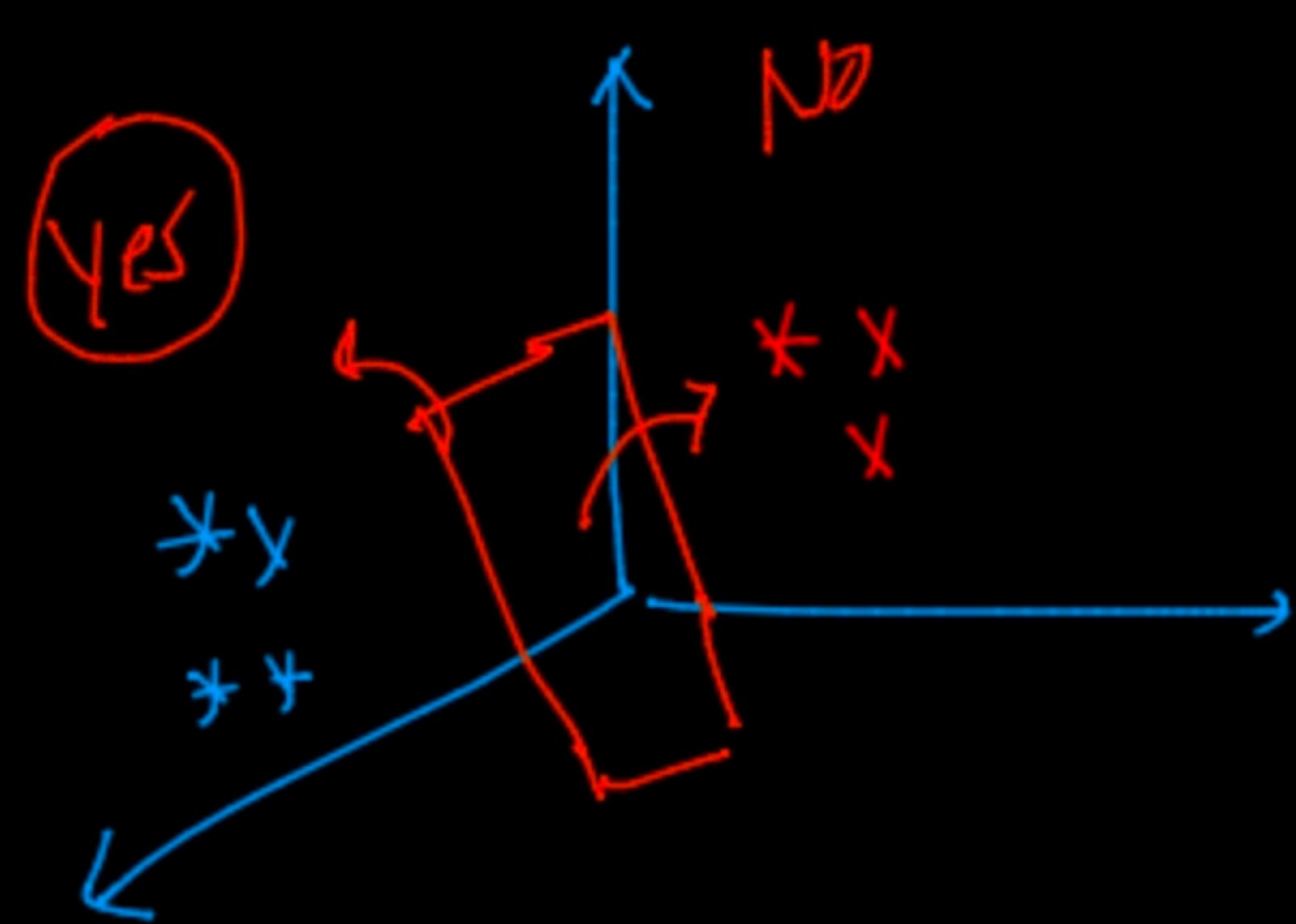
$w_1x_1 + w_2x_2 + w_0 = 0 \rightarrow \boxed{2D} \text{ line}$



3D

line
(2D)
Plane

ML
logistic



50% 50%
↑ ↑

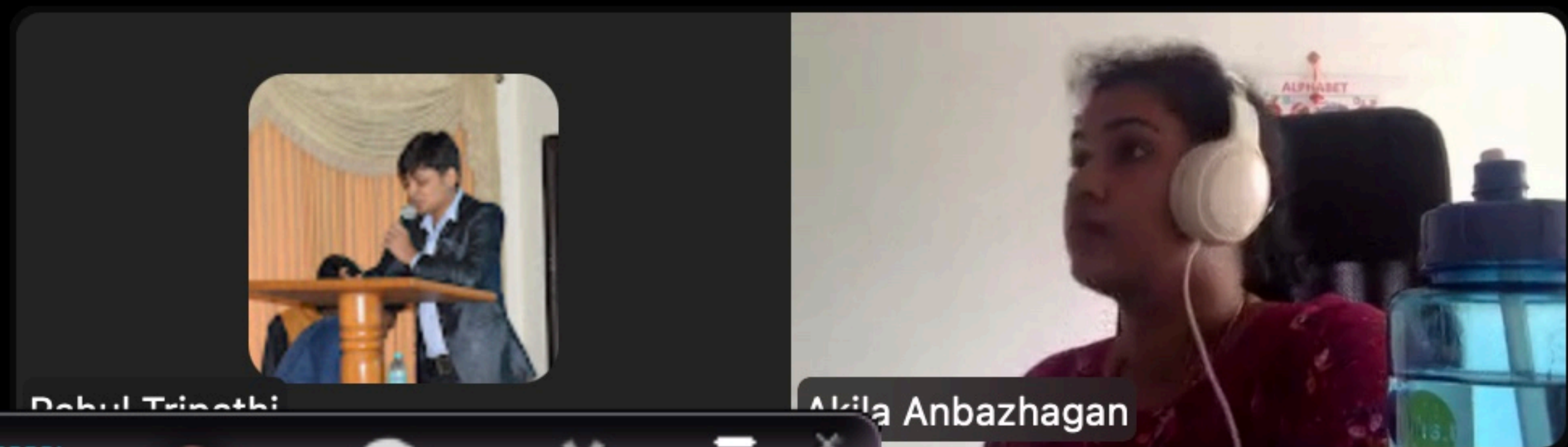
ML

{ }
ML math

$$w_1 x_1 + w_2 x_2 + w_3 x_3 + w_0 = 0 \quad \text{--- -- -- --} \rightarrow \boxed{nD}$$

Hyperplane

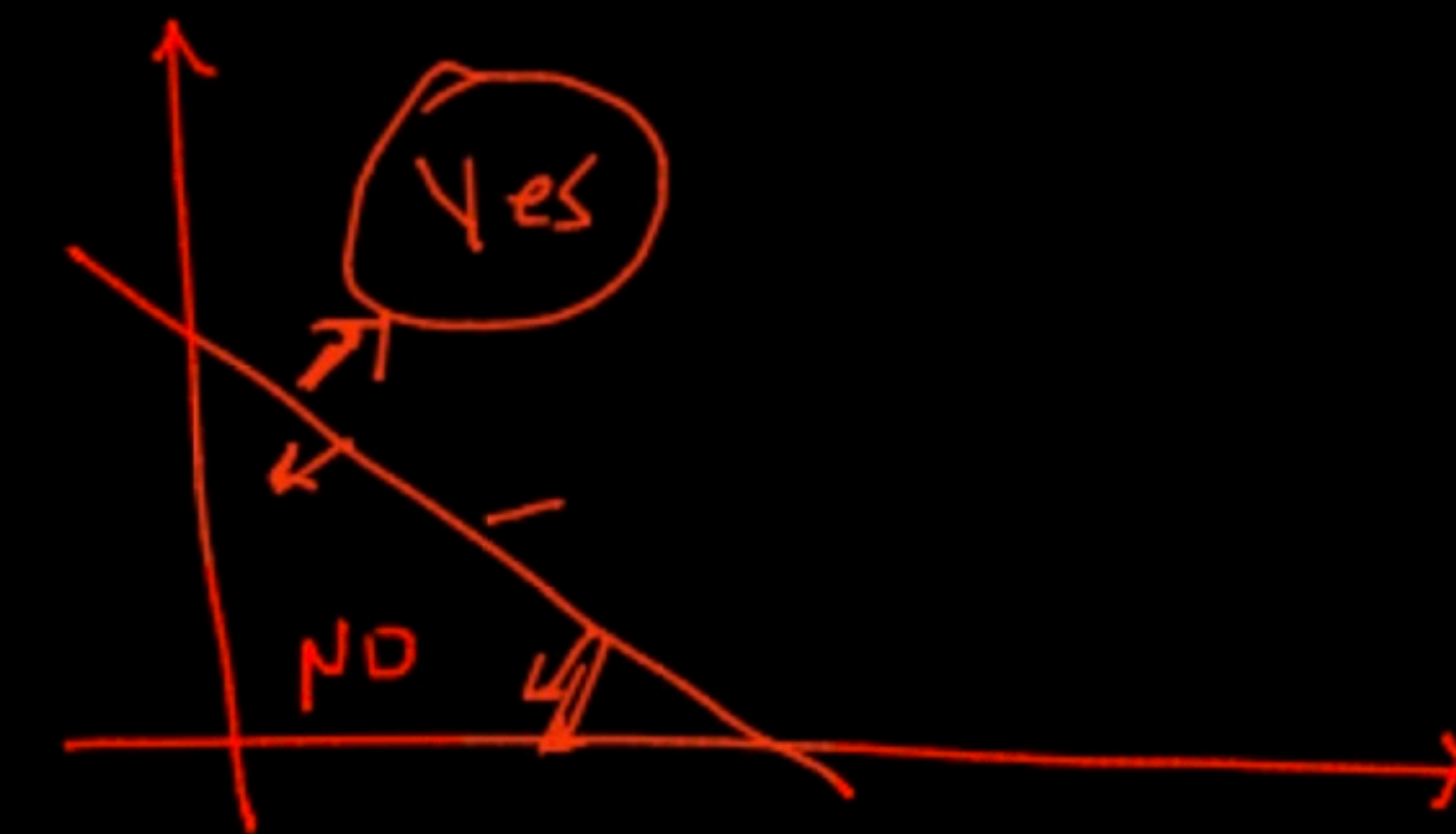
$$w_1 x_1 + w_2 x_2 + w_3 x_3 + w_0 = 0$$



$$w_0 + \sum_{i=1}^n w_i x_i = 0$$

→ new
↳ Hyperplane

constant



$$w_1 = 0.7$$

$$w_2 = 0.8$$

$$w_0 = 1$$

$$\begin{bmatrix} w_1 \\ w_2 \end{bmatrix} [10, 15]$$

New-customer $\rightarrow [10, 150]$

'Rule'

$$w_1 \times 10 + w_2 \times 150 + 1 = 0$$

$$0.7 \times 10 + 0.8 \times 150 + 1$$

$$\downarrow$$

$$= 127$$

$$w_0 + \sum_{i=1}^n w_i x_i > 0$$

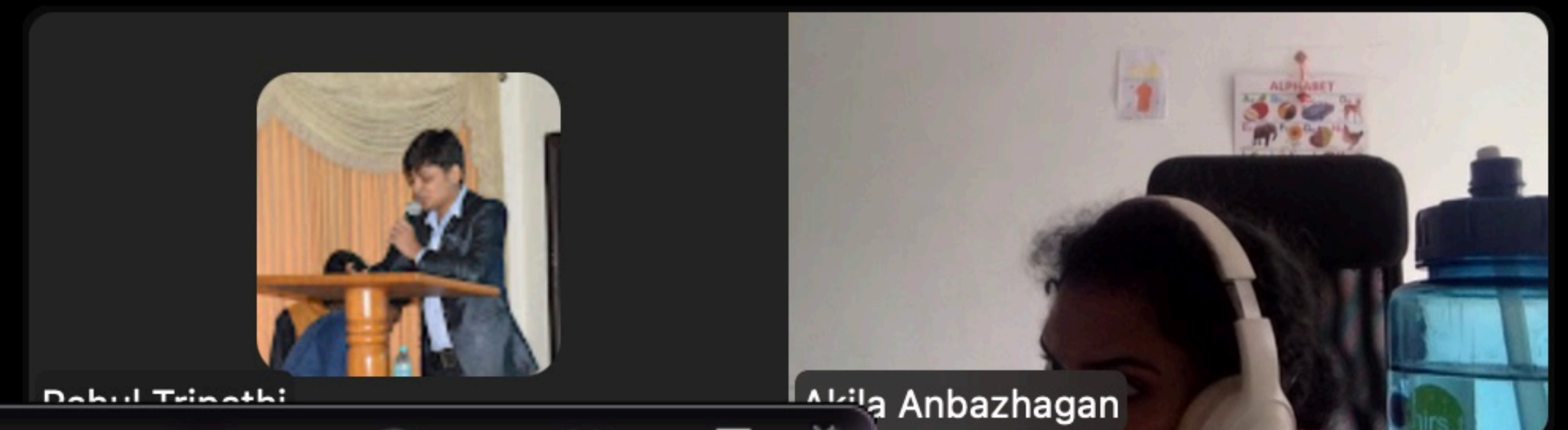
Yes

$$< 0$$

No

$+w$

$-w$



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